

Mental Health and the Metaverse

Ample Opportunities or Alarming Threats for Mental Health in Immersive Worlds?

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ABSTRACT

The proportion of the global burden of disease attributable to mental disorders has been continuously increasing since its first recording and is already at a critical level. While responses to the global mental health crisis are urgently required, implementation of countermeasures has so far been inadequate. Due to potential cost-effectiveness and scalability, technological solutions might play an important role in addressing current and projected mental health needs. The various proposed applications of the metaverse might provide unique opportunities in this regard, while also potentially creating additional mental health strain on individuals. In this panel, we will discuss the challenges and opportunities the metaverse might pose for the future of mental health, and the steps we can take now to harness the metaverse for positive change in global mental health – or prevent it from exacerbating already pressing concerns.

CCS CONCEPTS

• **Applied computing** → Life and medical sciences; Consumer health; • **Human-centered computing** → Human computer interaction (HCI); Interaction paradigms; Mixed / augmented reality; Human computer interaction (HCI); Interaction paradigms; Virtual reality.

KEYWORDS

Augmented Reality (AR), Extended Reality (XR), Mental Health, Metaverse, Mixed Reality, Virtual Reality (VR)

ACM Reference Format:

Gunther Meinlschmidt, Stefanie Herta, Stefan Germann, Cliona Chee Pui Khei, Sebastian Klöss, and Moritz Borrmann. 2023. Mental Health and the Metaverse: Ample Opportunities or Alarming Threats for Mental Health in Immersive Worlds?. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*, April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3544549.3583750>

1 PANEL PARTICIPANTS

1.1 Name of moderator:

- Dr. Stefan German (he/him), CEO, Botnar Foundation, Basel, Switzerland
- ⇒ Contributing a perspective of a large Foundation active in the field of digital and mental health and longstanding expertise in mental healthcare innovations.

1.2 Name of panelists:

- Cliona Chee Pui Khei (she/her), Candidate of B. Eng. Biomedical, Department of Biomedical Engineering & Health Sciences Faculty of Electrical Engineering, UTM Johor, Malaysia
- ⇒ Contributing a junior biomedical engineering, mental health, and start-up perspective.
- Dr. Sebastian Klöss (he/him), Division Manager of Consumer Technology, AR/VR & Metaverse, Germany's Digital Association Bitkom, Berlin, Germany
- ⇒ Contributing an IT, telecommunications, and new media industry perspective.

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CHI EA '23, April 23–28, 2023, Hamburg, Germany

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ACM ISBN 978-1-4503-9422-2/23/04.

<https://doi.org/10.1145/3544549.3583750>

- Dr. Gunther Meinschmidt (he/him), Professor of Clinical Psychology and Psychotherapy, Psychotherapist, IPU Berlin & University Hospital and University of Basel, Germany & Switzerland
- ⇒ Contributing a mental health care, research, and start-up perspective.
- Dr. Kremlin Wickramasinghe (he/him), Acting Head of the WHO European Office for the Prevention and Control of Noncommunicable Diseases (NCD Office), Geneva, Switzerland
- ⇒ Contributing a global public (mental) health perspective.

1.3 Interactive content moderator and backup panelists:

- Moritz Borrmann (he/him), M.A., IPU Berlin, Germany
- ⇒ Contributing a junior mental healthcare, research, and tech-use perspective.
- Stefanie Herta (she/her), B.Sc., IPU Berlin, Germany
- ⇒ Contributing a junior mental healthcare, Web-3, and research perspective.

2 INTRODUCTION

In 2022, the World Health Organization's (WHO) world mental health report highlighted that approximately 970 million individuals globally suffered from mental health disorders in the year 2019; approximately 60% of these individuals suffered from depression and anxiety in particular [2, 3]. Suicide continues to remain a major cause of death among young adults worldwide, with over 700,000 deaths in 2019 alone. Furthermore, mental disorders stand as one of the leading reasons for years lived with disability. Mental health conditions can also have a substantial impact on an individual's

social life, the social costs of which are difficult to quantify and include in economic cost analyses [4].

Apart from ongoing and widespread suffering, mortality, and social costs due to mental health conditions, economic implications are vast, with an estimated cost of 2.5 trillion USD per annum. Within this estimate, approximately 1.7 trillion USD occurred in productivity losses, which notably outweighed the 0.8 trillion USD sum of direct costs of care [5]. The projected total cost is said to rise to 6 trillion USD by the year 2030. Furthermore, there are substantial economic costs concerning the value linked to the participation of individuals in the unpaid labor force, for example as homemakers and caretakers [4].

Despite the high prevalence of mental disorders and their associated suffering and costs, mental disorders remain largely undertreated and undiagnosed [3]. Furthermore, afflicted individuals are met with mental health care systems that are under-resourced, characterized by major gaps in technologies for mental health care [3].

Over a decade ago, research priorities were set by the WHO and others to improve the lives of those with mental health conditions. Of particular note is the priority given to the development of mobile and information technologies, with the aim of improving access to evidence-based care [6]. In alignment with the WHO's research priority, the Lancet Commission on Global Mental Health and Sustainable Development stressed the value of scaling up mental healthcare [7]. In line with this, the WHO recently emphasized the potential for digital technologies to aid national efforts in providing universal access to mental health care, stressing that digital technologies can help provide greater flexibility and convenience for individuals, reducing travel costs and time, and helping to avoid stigma through anonymity [3]. Notably, compelling evidence suggests that digital approaches can be effective in supporting mental

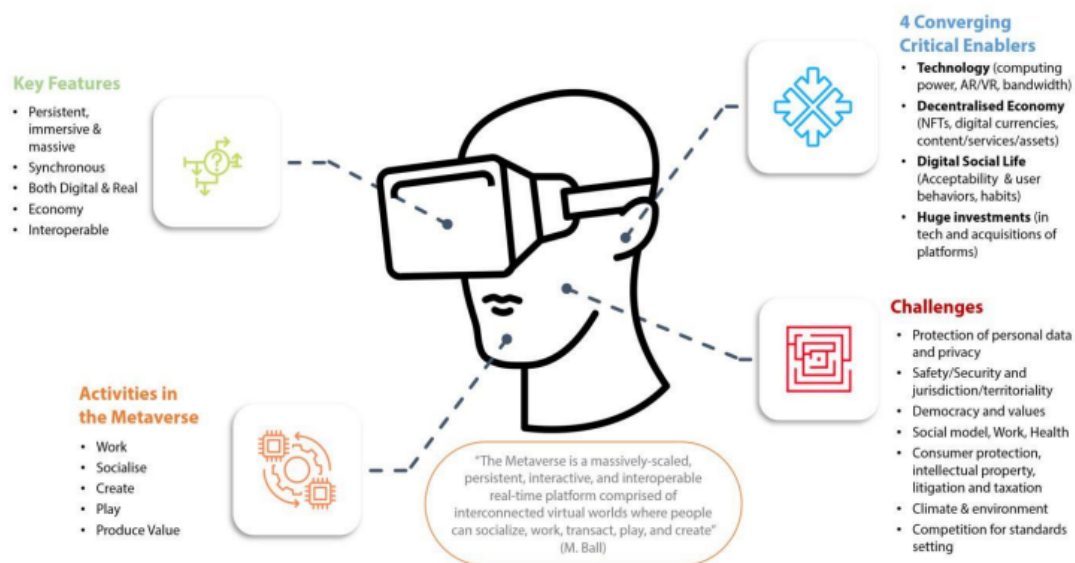


Figure 1: Potential key features, usage possibilities, and challenges of the metaverse Notes: Source: [1], Abbrev.: AR, augmented reality; VR, virtual reality; NFTs, Non-Fungible Tokens. ©Analysis and Research Team (ART) of the European Council.

health [3]. However, despite this existing evidence, the uptake and integration of digital technologies in the mental healthcare system has been inconsistent, often limited, slow, or even virtually non-existent [8].

Important questions remain as to how new and emerging technologies, such as the metaverse, could contribute to the future of mental health care, how potential mental health risks may arise through using them, and how we can safeguard mental health in this space. While views on the further uptake of the metaverse are rather diverse, it seems reasonable to anticipate that the metaverse may develop into a powerful new digital space with ample relevance for mental health.

3 DEPICTING THE METAVERSE

There is no single and commonly accepted definition of the ‘metaverse’. It is generally perceived to be a collective and shareable virtual space that is i) not exclusively focused on a single set of goals, such as gaming, ii) consistent, in that it allows for a synchronous experience in real-time for all who access it, iii) continuously present, in that it exists irrespective of whether one is currently using the metaverse, iv) created through the fusion of both virtually enhanced or augmented physical reality with digital reality (virtual space) [9] and has v) a high level of interoperability with decentralization [9].

In recent years, there has been a great deal of excitement about the metaverse, partially instigated by big technology players who have pledged to invest substantial resources [10]. However, there are greatly diverging predictions for the future of the metaverse, ranging from predictions of a merely short-lived interest in that may evaporate within a mere few years, to predictions that the metaverse will develop into an omnipresent technology, paired with the viewpoint that the metaverse “could constitute an entirely new space offering limitless possibilities and the potential to change our lives,” see 1 [9].

Regarding technological capabilities, over the past decades, extended reality (XR) systems have become more immersive and complex, with features such as the ability for multiple users to be immersed in the same scenery, refined avatar-based representations of users, and real-time visualization of facial and other body movements. Beyond XR, other technologies, such as blockchain have been explored to constitute and enrich the metaverse. These developments have been paving the way for the emergence of the metaverse. However, their implications and applications for mental health - specifically in the context of the metaverse - are just beginning to be explored.

4 THE METAVERSE AND MENTAL HEALTH

The metaverse could become relevant for mental health in two key ways. Firstly, the metaverse could offer a space to reach out and deliver mental health care to individuals in need. Secondly, it may also pose potential mental health risks. Therefore, minimizing potential exposure to these risks, while supporting those already affected by mental challenges, will be a key priority. Simultaneously, there is an urgent need to identify key mechanisms and procedures for mental health care delivery in the metaverse.

With regard to mental health conditions related to the metaverse, one may distinguish between four conceptually distinct categories of concern:

1. Mental health conditions that are primarily emerging from analog behavior and living situations and are cared for in analog settings.
2. Mental health conditions that are primarily emerging from analog behavior and living situations and are cared for in the metaverse.
3. Mental health conditions that are primarily emerging from behavior and situations in the metaverse and cared for in analog settings.
4. Mental health conditions that are primarily emerging from behavior and situations in the metaverse and are cared for in the metaverse.

While these four prototypic concepts serve as a framework for mental health and the metaverse, with the latter three being discussed in more detail below, it should be noted that the boundaries between categories are sliding in individual cases. For example, mental health interventions may combine i) therapy in the ‘real world’ that is within an analog setting, ii) augmented reality (AR), where computer-generated data are fused into analog environments, and iii) virtual space, where a computer has constructed an environment in its entirety [11] (cf., the mixed-reality continuum [12]).

5 SAFEGUARDING MENTAL HEALTH IN THE METAVERSE

Metaverse use is expected to be paired with several mental health risks. Deduced from evidence from other social media and digital technology use, these may include attention deficit attention-deficit symptoms, disrupted sleep, or hindered brain development [9, 13], as well as technology addiction (such as gaming disorder), and social isolation [14].

Furthermore, the metaverse may elicit mental health challenges that go beyond risks that have been identified relating to the use of well-established digital technologies; these include:

1. The use of the metaverse may increase the risk of maladaptive escapism, due to its immersive power [15].
2. Offensive and sexual assaults are already a problem in current technology usage, and the immersive power of the metaverse may exacerbate this issue, potentially giving rise to disorders related to traumatic events which occurred in the metaverse [16].
3. The metaverse may lead to new mental health challenges, such as fuzzy and incongruent identities, dissociation, and new forms of mental health concerns relating to avatar body image [15].
4. Dissociative experiences may occur due to XR technology placing demands on the brain to create and maintain two separate models of the world [17]. These experiences may also play an important role in bodily discomfort associated with XR (“cybersickness”) [18].

6 CALL FOR ACTION TO MAINSTREAM MENTAL HEALTH IN THE METAVERSE

To protect mental health in the metaverse, the following actions may be considered:

- Prevent actions that harm mental health, including physical, emotional, and sexual violence, aggression, and discrimination.
- Identify and alert users to unhealthy usage patterns, using indicators such as duration.
- Notify and inform users about emotionally demanding circumstances such as “ghosting” or “cyberbullying”.
- Educate users on using the metaverse sensibly, and encourage metaverse literacy by offering guidance and training, for example, encouraging and supporting people to maintain a healthy ratio between time spent in the metaverse and the physical world, while focusing on groups that may be especially vulnerable to the negative effects of using the metaverse, including children and adolescents.
- Identify and alert of any pre-existing factors, such as media addiction, which may increase the risk of negative effects of metaverse use.
- Provide accessible and effective evidence-based support for mental health conditions of varying intensity levels, like in the physical world.
- Ensure accessibility and safeguarding of such support for particularly underserved and vulnerable populations.
- Ensuring that such support services adhere to the most recent ethical and data protection standards.
- Prevent unclear and inaccurate identities: There will be a need for trusted digital identities for people, goods, and organizations that can be used across different platforms, especially in the context of mental healthcare.
- Cultivate a culture of support and mutual respect within the metaverse.
- Enable social interactions that go beyond just virtual, to augmented and to physical reality.
- Ensure that commercial organizations are held accountable for any negative consequences derived from using their metaverse-related services.
- Uphold employers’ responsibility when staff members use the metaverse.

7 FUTURE ITERATIONS OF THE METAVERSE

Various technologies are about to emerge that facilitate the interaction and communication of humans with the real and virtual environment, including brain-computer interfaces, gesture recognition, omnidirectional treadmills, and speech recognition. In the future, the VAKOG-metaverse (short for visual, acoustic, kinesthetic, olfactory, gustatory) and brain-computer interfaces may provide a more “immersive internet of senses” with the ability to experience smells, touch, and taste. It is difficult to predict the exact trajectory of technology, but it seems reasonable to assume that the long-term effects of future introductions of a full set of such technologies will have far-reaching consequences. Taken together, it should be seriously considered to act now to ensure that the metaverse and subsequent future iterations are a safe place for mental health, and

to realize its potential to help those with mental health needs and prevent potential harm.

8 PANEL FORMAT, SESSION PLAN, AND INTERACTIVE ELEMENTS

The metaverse is a constantly evolving, largely prospective construct with comparatively few observable data, and this may be a contributing factor to the uncertainty and conceptual indeterminacy surrounding the metaverse [19]. The same is necessarily true for the topic of how mental health relates to the metaverse. Hence, we believe that a panel with strong interactive components at a major event like CHI23 is an ideal setting to exchange thoughts and ideas about the current state of the metaverse and to contribute to shaping its future, safeguarding, and fostering mental health. To this end, we believe that designing a hybrid participatory framework, with a strong focus on audience interaction, for our panel discussion provides a unique opportunity. As such, following the panel discussion and as a showcase, we will optionally invite those interested audience members and panelists to continue an informal exchange within a 2D and 3D accessible space in the metaverse.

Panelists unable to attend in person will attend the panel via video conferencing. Their video will be projected into the conference room and they in turn will be provided with a live audio-visual feed of the conference room. Both panelists and in-place audience members shall be advised that the panel discussion will be transmitted.

The outline of the panel’s session plan is as follows:

1. Welcome and opening lines by the first author (duration: 5’).
2. Introduction of, and initial round of position statements by panelists (duration: 15’).
3. Moderated discussion between panelists, including audience interaction via live chat (duration: 45’) and after a while incorporating voices from in-place attendances, according to the following three topics:
 - a. What are the challenges for mental health in the metaverse?
 - b. What are the opportunities for mental health care in the metaverse?
 - c. What are potential steps and what actions should be taken to make the metaverse a place that is safe and supportive for mental health?
4. Wrap-up and take-away statements by panelists (duration: 5’).
5. Farewell and closing lines (duration 5’)
6. Invitation to optionally attend post-session informal exchange in metaverse instance (duration: 15’–30’).

Audience interaction will be implemented as a continuously running live chat, providing audience members with the opportunity to engage in interactions with both panelists as well as other audience members via chat messages and poll votes. Rapidly expanding scientific inquiry into live chat interactions, such as commonly found on platforms such as Twitch or YouTube Live, indicates its potential to sustainably engage viewers, positively affect their experience, and foster closer relationships amongst audience members, as well as between audience members and content creators [20]. In addition, audience members will be informed of the option to raise their

hand during the moderated discussion part of the panel. They will then have the opportunity to raise points or ask questions upon invitation by the panel moderator.

The live chat shall be visible for audience members both physically present at the venue (via projector and screen) and participating digitally (via virtual access). Panelists are invited to integrate their reactions to audience messages and votes in their contributions to the panel discussion as they see fit. A live chat moderator will bring particularly salient contributions to the attention of the panelists, if necessary, as well as ensure compliance in chat behavior.

Following the panel discussion, we will host a metaverse instance for all interested audience members, allowing all audience members to explore the metaverse, as well as engage in ad hoc small-scale interactions. The instance will be opened after the panel discussion for an informal gathering of interested audience members and panelists and may ideally serve as a place to trigger further activities toward the goal of ensuring that the metaverse is a safe and positive place for mental health.

ACKNOWLEDGMENTS

GM received funding from the Stanley Thomas Johnson Stiftung & Gottfried und Julia Bangerter-Rhyner-Stiftung under projects no. PC 28/17 and PC 05/18, from Gesundheitsförderung Schweiz under project no. 18.191/K50001, from Swiss Heart Foundation under project no. FF21101, from the Research Foundation of the International Psychoanalytic University (IPU) Berlin, from the Swiss National Science Foundation (SNSF) under project no. 100014_135328, from the Hasler Stiftung under project no. 23004, and in the context of a Horizon Europe project from the Swiss State Secretariat for Education, Research and Innovation (SERI) under contract number 22.00094. Further, GM receives funding from Wings Health Inc. in the context of a proof-of-concept study. GM is co-founder, member of the board, and holds stock in Therayou AG, which is active in the field of digital and blended mental healthcare. SG is co-founder, member of the board, and holds stock in Therayou AG, which is active in the field of digital and blended mental healthcare. SK is employed by Germany's Digital Association Bitkom which represents the interests of its >2000 members.

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